STL Dust Simulations for the BICEP/Keck Foreground Model Suite

1 Scientific Context

The BICEP/Keck program constrains the tensor-to-scalar ratio r from degree-scale CMB B-mode polarization. At the sensitivity of BK18 and the upcoming BK24 analysis, the dominant astrophysical systematic is Galactic polarized foreground emission, especially dust. The BICEP/Keck likelihood fits multi-frequency BB spectra with a model containing lensed ΛCDM, tensor modes, dust, synchrotron, and noise. In BK18, the baseline dust model was a power-law angular spectrum and a modified-blackbody spectral energy distribution (SED), with fitted dust amplitude

$$A_{d,353} \simeq 4.4\,\mu\mathrm{K}^2$$

at $\ell = 80$ and 353 GHz on the BICEP/Keck field.

The requested deliverable is not a deterministic reconstruction of the true BICEP sky. The requested deliverable is an ensemble of independent foreground realizations whose statistics are realistic for the BICEP field and consistent with the existing BICEP/Keck constraints. These simulations will be passed through the BICEP/Keck analysis machinery to test whether foreground complexity not captured by the baseline parametric model can bias the recovered value of r.

2 Target Data Product

The target product is an ensemble of synthesized foreground maps

$$\left\{ \tilde{\boldsymbol{s}}_\nu^{\prime(a)} \right\}_{a=1}^{N_{\mathrm{sim}}}, \qquad \tilde{\boldsymbol{s}}_\nu^{\prime(a)} = \left(\tilde{I}_\nu^{\prime(a)}, \tilde{Q}_\nu^{\prime(a)}, \tilde{U}_\nu^{\prime(a)} \right),$$

on the BICEP/Keck sky patch, at the observing bandpasses supplied by BICEP/Keck. The desired ensemble size is

$$N_{\mathrm{sim}} \simeq 500,$$

matching the standard BICEP/Keck simulation count. The maps should be provided at

$$N_{\mathrm{side}} = 512,$$

and should have controlled angular statistics over the approximate range

$$30 \lesssim \ell \lesssim 500.$$

If necessary for a first delivery, the upper end may be reduced to $\ell \simeq 300$, but the degree-scale range around $\ell \simeq 80$ is essential.

The primary foreground component is polarized thermal dust. Synchrotron may also be included if a corresponding amplitude/SED prescription is agreed upon, but the immediate problem is to generate dust realizations whose amplitude, power spectrum, non-Gaussian morphology, and frequency decorrelation are compatible with BICEP/Keck data.

3 Input Data and Hypotheses

Let $\Omega_{\rm HL}$ denote a collection of clean high-latitude sky patches outside the Galactic plane. The reference data are cleaned Planck polarized dust maps at 353 GHz,

$$\tilde{\boldsymbol{s}}_{353,p} = \left(\tilde{Q}_{353,p}, \tilde{U}_{353,p}\right), \qquad p \in \Omega_{\rm HL},$$

possibly together with an intensity dust tracer such as

$$\tilde{I}_{857,p}$$

or another high signal-to-noise dust morphology template. These are not "training" data in the machine-learning sense; they are the observed, cleaned Planck dust patches from which we measure the target statistics.

The working hypotheses are:

(H1) High-latitude Planck dust patches contain representative samples of the non-Gaussian polarized dust morphology relevant for the BICEP/Keck field.

(H2) A finite set of scattering-transform statistics, supplemented by explicit power-spectrum constraints, captures the multiscale, anisotropic, non-Gaussian statistics needed for BICEP/Keck foreground validation.

(H3) The generated realizations need not be phase-locked to the actual Planck sky at low multipoles. In fact, BICEP/Keck prefers statistically independent realizations rather than maps deterministically tied to "Planck truth" on the BICEP field. In other words, the goal is not to reproduce the particular observed sky realization pixel by pixel, but to reproduce the relevant ensemble statistics.

(H4) The high-latitude maps must be amplitude-calibrated using a simple map-space normalization from the component-separated BICEP patch, and then spectrally extrapolated so that, on the BICEP/Keck mask, their amplitudes and cross-frequency decorrelation are compatible with the BK18 constraints. Cross-frequency decorrelation means that the dust pattern at two frequencies is not exactly the same map multiplied by two constants; spatial variations of the dust SED can suppress the cross-spectrum between frequencies relative to their auto-spectra.

4 STL Statistical Model at 353 GHz

Let Φ denote the STL feature map. In the current Planck component separation code, Φ includes mean and variance, scattering covariance coefficients

$$S_1,\ S_2,\ S_3,\ S_4,$$

and, when enabled, radially binned Fourier power spectra. The statistics may include selected cross-statistics between channels, for example between polarized maps and an auxiliary intensity tracer.

The generative problem at 353 GHz can be written as follows. We seek a probability law $\mathcal{P}_{\rm dust}$ on synthesized BICEP-patch polarization maps

$$\tilde{\boldsymbol{s}}_{353}^{\prime(a)} = \left(\tilde{Q}_{353}^{\prime(a)}, \tilde{U}_{353}^{\prime(a)}\right)$$

such that its expected STL statistics match those measured from the high-latitude Planck reference data:

$$\mathbb{E}_{\tilde{\boldsymbol{s}}' \sim \mathcal{P}_{\rm dust}}\left[\Phi(\tilde{\boldsymbol{s}}', \tilde{I}'_{\rm aux})\right] \simeq \frac{1}{N_{\rm HL}} \sum_{p \in \Omega_{\rm HL}} \Phi(\tilde{\boldsymbol{s}}_{353,p}, \tilde{I}_{{\rm aux},p}).$$

The generated maps are then checked against the BICEP/Keck two-point constraints,

$$D_\ell^{BB,353} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{BB,353} \simeq A_{d,353} \left(\frac{\ell}{80}\right)^{\alpha_d},$$

but this is not intended to be a hard per-ℓ constraint in the synthesis objective. The primary calibration is instead a simple map-space amplitude renormalization, using the component-separated BICEP patch as the target amplitude reference.

Operationally, one may generate maps by solving, for each realization a, an optimization problem of the form

$$\tilde{\boldsymbol{s}}_{353}^{\prime(a)} = \arg\min_{\boldsymbol{u}} \left\| \Phi(\boldsymbol{u}, \tilde{I}_{\mathrm{aux}}^{\prime(a)}) - \widehat{\Phi}_{\mathrm{HL}} \right\|_2^2,$$

where $\widehat{\Phi}_{\mathrm{HL}}$ is the empirical STL summary of the high-latitude Planck reference data, after the chosen map-space amplitude normalization. Random initial conditions, random reference patch draws, or minibatch statistics provide independent realizations.

5 Frequency Scaling and Decorrelation

BICEP/Keck does not only need 353 GHz maps. It needs maps integrated over its observing bandpasses. A minimal dust SED model is a modified blackbody:

$$\tilde{Q}_\nu'(\hat{n}) = \tilde{Q}_{353}'(\hat{n})\, f_d(\nu;\beta_d,T_d), \qquad \tilde{U}_\nu'(\hat{n}) = \tilde{U}_{353}'(\hat{n})\, f_d(\nu;\beta_d,T_d),$$

where

$$f_d(\nu;\beta_d,T_d) = \frac{g_{\mathrm{CMB}}(353)}{g_{\mathrm{CMB}}(\nu)} \left(\frac{\nu}{353\,\mathrm{GHz}}\right)^{\beta_d} \frac{B_\nu(T_d)}{B_{353}(T_d)}.$$

This single-SED model produces no cross-frequency decorrelation: each frequency map has exactly the same morphology, only rescaled. A minimal decorrelating example is

$$\tilde{Q}_\nu'(\hat{n}) = \tilde{Q}_{353}'(\hat{n}) \left(\frac{\nu}{353\,\mathrm{GHz}}\right)^{\beta_d(\hat{n})}.$$

If $\beta_d(\hat{n})$ varies across the sky, two pixels with the same 353 GHz amplitude may be rescaled differently at 150 GHz. The 150 GHz and 353 GHz maps are then no longer identical up to a global multiplicative factor.

The email exchange explicitly asks for enough SED complexity to reproduce the expected decorrelation constraints, so the proposed model should include at least one of the following extensions:

(S1) spatially varying spectral index and/or temperature, $\beta_d(\hat{n})$ and $T_d(\hat{n})$;

(S2) a multi-layer dust model,

$$\tilde{Q}_\nu'(\hat{n}) = \sum_{k=1}^{K} \tilde{Q}_{k,353}'(\hat{n})\, f_d(\nu;\beta_{d,k}(\hat{n}),T_{d,k}(\hat{n})),$$

with the same expression for $\tilde{U}_\nu'$;

(S3) stochastic small-scale SED fluctuations whose amplitude is tuned to produce an allowed level of decorrelation.

For two frequencies ν_1, ν_2, define the dust decorrelation coefficient on the BICEP/Keck mask by

$$\Delta_d(\nu_1, \nu_2; \ell) = \frac{C_\ell^{BB,d}(\nu_1 \times \nu_2)}{\sqrt{C_\ell^{BB,d}(\nu_1 \times \nu_1) C_\ell^{BB,d}(\nu_2 \times \nu_2)}}.$$

BK18 found that the likelihood for the decorrelation parameter peaks at unity, i.e. at no detected decorrelation, and Fig. 18 of BK18 provides the relevant constraint for the 217×353 GHz comparison. Therefore the SED model should be rich enough to explore decorrelation, but tuned so that

$$\Delta_d$$

remains compatible with the BICEP/Keck likelihood on the sky mask. Strongly decorrelated models are useful stress tests only if clearly labeled as such.

6 Ratio-Map SED Generative Strategy

Following the discussion with François Boulanger, François Levrier, and Sébastien, the preferred practical route is to separate the problem into two coupled generative pieces:

(R1) a generative model for the reference dust morphology, for instance at 353 GHz;

(R2) a generative model for frequency-ratio maps, which encode the spatial variation of the dust SED and therefore the possible decorrelation between channels.

The ratio model is intended to be learned from simulations of pairs of frequency channels. Direct ratios of Q or U are undesirable because these fields change sign and have many zero crossings. We therefore define the ratio on polarized intensity,

$$\tilde{P}_\nu(\hat{n}) = \left[\tilde{Q}_\nu^2(\hat{n}) + \tilde{U}_\nu^2(\hat{n})\right]^{1/2},$$

possibly after smoothing, masking, or debiasing if required. For two channels 353 and 217 GHz, the basic ratio field is

$$\rho_{353/217}(\hat{n}) = \mathcal{R}_\epsilon \left[\tilde{P}_{353}(\hat{n}), \tilde{P}_{217}(\hat{n})\right],$$

where $\mathcal{R}_\epsilon$ denotes the chosen regularized ratio operation. In the simplest implementation,

$$\rho_{353/217}(\hat{n}) = \frac{\tilde{P}_{353}(\hat{n})}{\tilde{P}_{217}(\hat{n}) + \epsilon_P},$$

with a small regularization or mask to avoid unstable ratios in low-signal pixels. An equivalent log-ratio parameterization,

$$\eta_{353/217}(\hat{n}) = \log \rho_{353/217}(\hat{n}),$$

may be preferable numerically because it keeps the generated ratio positive. The essential object is a spatial field that describes how the polarized dust amplitude changes between two frequencies.

Let Ψ denote the scattering feature map used for ratio fields. From an ensemble of simulated ratios one measures

$$\widehat{\Psi}_{353/217} = \frac{1}{N_{\rm sim}^{\rm ratio}} \sum_{b=1}^{N_{\rm sim}^{\rm ratio}} \Psi\left(\rho_{353/217}^{(b)}\right).$$

The ratio realizations used in the BICEP simulations are then generated by solving

$$\rho_{353/217}^{\prime(a)} = \arg\min_v \left\|\Psi(v) - \widehat{\Psi}_{353/217}\right\|_2^2 + \lambda_{\rm ratio}\, \mathcal{C}_{\rm ratio}(v),$$

where $\mathcal{C}_{\mathrm{ratio}}$ denotes any additional constraints imposed on the ratio maps, such as smoothness, allowed dynamic range, or a target decorrelation level.

The 353 GHz morphology is still generated from the high-latitude dust statistics. Let

$$\mu_{\mathrm{HL}} = \frac{1}{N_{\mathrm{HL}}} \sum_{p \in \Omega_{\mathrm{HL}}} \Phi(\tilde{\boldsymbol{s}}_{353,p}, \tilde{I}_{\mathrm{aux},p})$$

be the mean scattering summary over high-latitude sky patches. Before using this target for BICEP/Keck simulations, the high-latitude maps are multiplied by a single scalar chosen from the component-separated BICEP patch. For example, define a robust polarized-intensity amplitude

$$A_P[\tilde{\boldsymbol{s}}] = \mathrm{RMS}_{W_{\mathrm{BK}}} \left[\left(\tilde{Q}^2 + \tilde{U}^2 \right)^{1/2} \right],$$

with the same smoothing, masking, and apodization convention applied to the high-latitude patches and to the BICEP patch. If $\tilde{\boldsymbol{s}}_{353}^{\mathrm{BK,cs}}$ denotes the component-separated BICEP-patch dust solution, set

$$\alpha_{\mathrm{HL}\to\mathrm{BK}} = \frac{A_P[\tilde{\boldsymbol{s}}_{353}^{\mathrm{BK,cs}}]}{A_P[\tilde{\boldsymbol{s}}_{353}^{\mathrm{HL}}]}, \qquad \tilde{\boldsymbol{s}}_{353,p} \leftarrow \alpha_{\mathrm{HL}\to\mathrm{BK}}\, \tilde{\boldsymbol{s}}_{353,p}.$$

The BICEP-adapted scattering target is then

$$\mu_{\mathrm{HL}}^{\mathrm{BK}} = \frac{1}{N_{\mathrm{HL}}} \sum_{p \in \Omega_{\mathrm{HL}}} \Phi(\alpha_{\mathrm{HL}\to\mathrm{BK}} \tilde{\boldsymbol{s}}_{353,p}, \alpha_{\mathrm{HL}\to\mathrm{BK}} \tilde{I}_{\mathrm{aux},p}),$$

or the corresponding normalized version if the auxiliary channel is not to be rescaled. The reference maps are generated as

$$\tilde{\boldsymbol{s}}_{353}^{\prime(a)} = \arg\min_{\boldsymbol{u}} \left\| \Phi(\boldsymbol{u}, \tilde{I}_{\mathrm{aux}}^{\prime(a)}) - \mu_{\mathrm{HL}}^{\mathrm{BK}} \right\|_2^2,$$

where $\mu_{\mathrm{HL}}^{\mathrm{BK}}$ denotes the high-latitude scattering target renormalized to the BICEP/Keck amplitude. There is no direct BB-spectrum fitting term in this first implementation. For an even simpler first implementation, one does not necessarily need to generate the reference morphology from scratch: one can start directly from the already solved component-separation result on the BICEP patch itself and use it as the initial 353 GHz dust template to test the extrapolation pipeline.

Given a synthesized reference map and a synthesized ratio map, one obtains the lower-frequency polarized amplitude by inversion of the ratio:

$$\tilde{P}_{217}^{\prime(a)}(\hat{n}) = \frac{\tilde{P}_{353}^{\prime(a)}(\hat{n})}{\rho_{353/217}^{\prime(a)}(\hat{n})}.$$

The simplest extrapolation keeps the polarization angle of the reference map and rescales the polarized vector:

$$\begin{pmatrix} \tilde{Q}_{217}^{\prime(a)}(\hat{n}) \\ \tilde{U}_{217}^{\prime(a)}(\hat{n}) \end{pmatrix} = \frac{\tilde{P}_{217}^{\prime(a)}(\hat{n})}{\tilde{P}_{353}^{\prime(a)}(\hat{n}) + \epsilon_P} \begin{pmatrix} \tilde{Q}_{353}^{\prime(a)}(\hat{n}) \\ \tilde{U}_{353}^{\prime(a)}(\hat{n}) \end{pmatrix}.$$

More generally, for any target band ν one can learn or prescribe a ratio field $\rho_{353/\nu}$ and set

$$\tilde{P}_{\nu}^{\prime(a)}(\hat{n}) = \frac{\tilde{P}_{353}^{\prime(a)}(\hat{n})}{\rho_{353/\nu}^{\prime(a)}(\hat{n})},$$

with the same polarized-vector rescaling used to obtain $\tilde{Q}'^{(a)}_\nu$ and $\tilde{U}'^{(a)}_\nu$. This makes the SED extrapolation a generative problem for polarized-intensity ratio fields rather than only a parametric modified-blackbody scaling. The ratio statistics control how much the polarized amplitude changes between channels, and therefore contribute to the decorrelation coefficients measured on the BICEP/Keck mask.

The validation requirement is unchanged: after extrapolation to the relevant BICEP/Keck bandpasses, the maps must satisfy the amplitude, angular-spectrum, and decorrelation constraints. In particular, the ratio-generated maps should be checked through

$$\Delta_d(\nu_1, \nu_2; \ell) = \frac{C_\ell^{BB,d}(\nu_1 \times \nu_2)}{\sqrt{C_\ell^{BB,d}(\nu_1 \times \nu_1) C_\ell^{BB,d}(\nu_2 \times \nu_2)}}$$

on the BICEP/Keck mask and compared with the BK18/BK24 allowed range.

6.1 Working in Q, U while constraining BB

The generative variables in the proposed pipeline are polarization maps,

$$\tilde{\mathbf{s}}'^{(a)}_\nu = \left(\tilde{Q}'^{(a)}_\nu, \tilde{U}'^{(a)}_\nu\right),$$

or quantities derived from them such as the polarized intensity $\tilde{P}'_\nu = (\tilde{Q}'^2_\nu + \tilde{U}'^2_\nu)^{1/2}$. The BICEP/Keck foreground constraints, however, are naturally stated in terms of E/B spectra, in particular the dust BB amplitude at $\ell \simeq 80$ and the shape of D_ℓ^{BB} over the BICEP/Keck range.

The intended strategy is therefore not to microtune the BB spectrum mode by mode, nor to use BB as the primary amplitude calibration. The primary amplitude calibration is the single map-space factor $\alpha_{\mathrm{HL}\to\mathrm{BK}}$ defined from the component-separated BICEP patch. After this calibration, each synthesized Q, U realization is transformed to E, B on the BICEP/Keck patch, with the same mask/apodization convention used for the validation spectra, and its BB spectrum is measured:

$$\widehat{D}_\ell^{BB,(a)} = \widehat{D}_\ell^{BB}\left[W_{\mathrm{BK}}\left(\tilde{Q}'^{(a)}_\nu, \tilde{U}'^{(a)}_\nu\right)\right].$$

This measured spectrum is then used as a diagnostic only. It verifies that the map-space calibration gives a BB amplitude and slope compatible with BICEP/Keck, but it is not used to fit or overwrite the realization.

This distinction is important. The scattering model should determine the polarized morphology in Q, U, and the resulting BB spectrum should emerge from that morphology after the usual $Q/U \to E/B$ projection. The BB constraint is used to flag or reject realizations whose spectra are grossly incompatible with BICEP/Keck data. It should not be used as a hard per-ℓ target that overwrites the non-Gaussian structure or frequency-dependent morphology generated by the model.

7 Calibration and Validation Against BICEP/Keck Constraints

The operational calibration is deliberately simple. We first use the component-separated BICEP patch to define a single map-space amplitude factor, and we multiply the high-latitude Q, U maps by that number before measuring the BICEP-adapted scattering target. The BB spectrum is then computed only after synthesis and SED extrapolation, as a validation product.

At minimum, on the BICEP/Keck mask:

(C1) the map-space polarized-intensity amplitude at 353 GHz should match the component-separated BICEP-patch reference, through the single scalar $\alpha_{\mathrm{HL}\to\mathrm{BK}}$;

(C2) after this calibration, the resulting BB spectrum should be checked against the BK18 scale, approximately $A_{d,353} \simeq 4.4\,\mu\mathrm{K}^2$ at $\ell = 80$, or the updated BK24 target if supplied;

(C3) the angular spectrum should have an acceptable slope over $30 \lesssim \ell \lesssim 500$, but it is not microtuned bin by bin;

(C4) the cross-frequency decorrelation coefficient should lie inside the allowed BICEP/Keck range;

(C5) the maps should not be locked to the observed BICEP/Planck phases at low multipoles;

(C6) the ensemble scatter across realizations should be available, since BICEP/Keck will use the simulations to estimate bias and uncertainty in recovered r.

8 Mathematical Workflow and Problem Statement

The proposed workflow can be summarized as a sequence of maps and constraints. First, start from cleaned high-latitude Planck polarization patches

$$\left\{\tilde{\boldsymbol{s}}_{353,p} = (\tilde{Q}_{353,p}, \tilde{U}_{353,p})\right\}_{p\in\Omega_{\mathrm{HL}}}, \qquad \tilde{I}_{\mathrm{aux},p}.$$

Measure their empirical STL summary

$$\widehat{\Phi}_{\mathrm{HL}} = \frac{1}{N_{\mathrm{HL}}} \sum_{p\in\Omega_{\mathrm{HL}}} \Phi(\tilde{\boldsymbol{s}}_{353,p}, \tilde{I}_{\mathrm{aux},p}),$$

after applying the single map-space normalization. The normalization factor $\alpha_{\mathrm{HL}\to\mathrm{BK}}$ is inferred from the component-separated BICEP patch.

Second, synthesize independent BICEP-patch dust realizations at 353 GHz:

$$\tilde{\boldsymbol{s}}'^{(a)}_{353} = \arg\min_{\boldsymbol{u}} \left[\left\|\Phi(\boldsymbol{u}, \tilde{I}'^{(a)}_{\mathrm{aux}}) - \widehat{\Phi}_{\mathrm{HL}}\right\|_2^2\right].$$

The prime distinguishes synthesized maps from the cleaned Planck reference maps. The index a labels independent statistical realizations.

Third, validate the amplitude of the synthesized realization on the BICEP/Keck patch. If $W_{\mathrm{BK}}(\hat{n})$ is the BICEP/Keck mask or apodization window, measure both the map-space polarized-intensity amplitude and the induced BB amplitude:

$$\widehat{A}^{(a)}_{P,353} = A_P\left[W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{353}\right], \qquad \widehat{D}^{BB,(a)}_{\ell} = \widehat{D}^{BB}_{\ell}\left[W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{353}\right].$$

No BB-based rescaling is applied at this stage in the baseline workflow. The BB spectrum is kept as a validation diagnostic for compatibility with BICEP/Keck.

Fourth, apply an SED model to obtain bandpass maps

$$\left\{\tilde{\boldsymbol{s}}'^{(a)}_{\nu}\right\}_{\nu\in\mathcal{B}},$$

where $\mathcal{B}$ is the set of BICEP/Keck observing bandpasses. The SED parameters must be tuned so that the decorrelation coefficients

$$\Delta^{(a)}_{d}(\nu_1, \nu_2; \ell) = \frac{C^{BB,d}_{\ell}\left[W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{\nu_1} \times W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{\nu_2}\right]}{\sqrt{C^{BB,d}_{\ell}\left[W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{\nu_1} \times W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{\nu_1}\right] C^{BB,d}_{\ell}\left[W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{\nu_2} \times W_{\mathrm{BK}}\tilde{\boldsymbol{s}}'^{(a)}_{\nu_2}\right]}}$$

remain compatible with the BK18/BK24 allowed range.

In compact form, the mathematical problem is therefore to construct an ensemble distribution $\mathcal{P}_\Theta$ for

$$\{\tilde{\boldsymbol{s}}'_\nu\}_{\nu\in\mathcal{B}}$$

such that, on the BICEP/Keck patch,

$$\mathbb{E}_{\mathcal{P}_\Theta}\left[\Phi(\tilde{\boldsymbol{s}}'_{353}, \tilde{I}'_{\rm aux})\right] \approx \widehat{\Phi}_{\rm HL},$$

$$A_P[\tilde{\boldsymbol{s}}'_{353}] \approx A_P[\tilde{\boldsymbol{s}}^{\rm BK,cs}_{353}],$$

with the amplitude matching imposed through the single map-space factor $\alpha_{\rm HL\to BK}$. The resulting BB spectrum is then checked a posteriori,

$$D_\ell^{BB,d} \text{ compatible with } A^{\rm BK}_{d,353}\left(\frac{\ell}{80}\right)^{\alpha_d}, \qquad 30 \lesssim \ell \lesssim 500,$$

but it is not fitted directly in the synthesis objective. Finally,

$$\Delta_d(\nu_1, \nu_2; \ell) \in \mathcal{C}_{\rm BK},$$

where $\mathcal{C}_{\rm BK}$ denotes the BICEP/Keck decorrelation constraint. The scientific test is then to run the BICEP/Keck likelihood on

$$\mathrm{CMB}^{(a)} + \mathrm{noise}^{(a)} + \tilde{\boldsymbol{s}}'^{(a)}_\nu$$

and measure the induced shift

$$\Delta r = \widehat{r}_{\rm recovered} - r_{\rm input}.$$

The model is successful if it provides foreground realizations that are realistic, independent, BICEP-compatible, and useful for quantifying whether non-Gaussian dust morphology or SED complexity can bias r at a relevant level.